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Operating system

Lab Task 6

Task:1

a)Parent writes numbers instead of string

The Parent sends a number , but the Child is expecting text.

So Child prints garbage and weird symbols because it's expected a string.

b)Write longer string than buffer size

The message gets cut off at 100 characters size of buffer and read() only reads the buffer size , So there is no stop (null terminator), the child may print random garbage following the text

Task:2

Code Snapshot:

```
#include <stdio.h>
#include <unistd.h>
#include <string.h>

int main() {

    int fd[2];
    pid_t pid;
    char write_msg[50];
    char read_msg[50];

    pipe(fd);

    pid = fork();

    if (pid > 0) {
        //parent process...
        close(fd[0]);

        printf("Enter a string: ");
        scanf("%s", write_msg);
        write(fd[1], write_msg, strlen(write_msg) + 1);
        close(fd[1]);
    }
    else {
        // child process
        close(fd[1]);
        read(fd[0], read_msg, sizeof(read_msg));

        int length = strlen(read_msg);
        printf("Reverse String is : ");
        for(int i = length - 1; i >= 0; i--) {
            printf("%c", read_msg[i]);
        }
        printf("\n");

        close(fd[0]);
    }

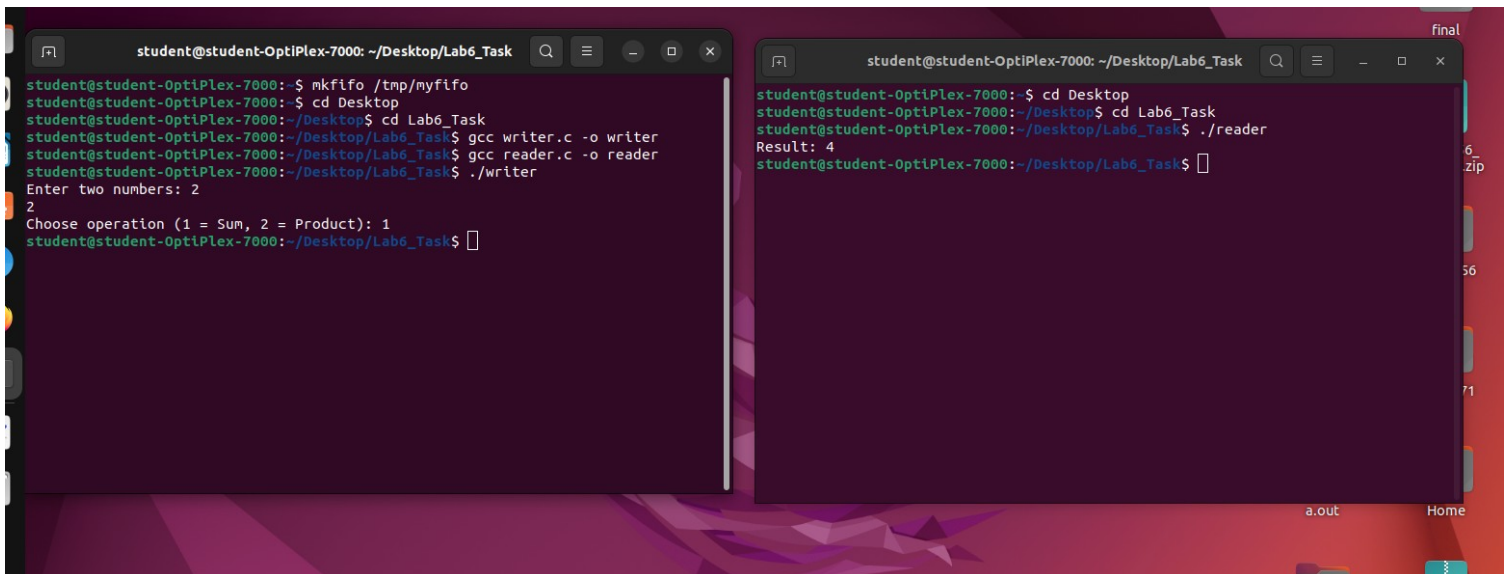
    return 0;
}
```

Output:

```
student@student-OptiPlex-7000:~/Desktop$ gcc ordinary_pipe.c
student@student-OptiPlex-7000:~/Desktop$ ./a.out
Enter a string: fahad
Reverse String is : dahaf
student@student-OptiPlex-7000:~/Desktop$
```

Task 3:

Output:



```
student@student-OptiPlex-7000: ~/Desktop/Lab6_Task
student@student-OptiPlex-7000:~$ mkfifo /tmp/myfifo
student@student-OptiPlex-7000:~$ cd Desktop
student@student-OptiPlex-7000:~/Desktop$ cd Lab6_Task
student@student-OptiPlex-7000:~/Desktop/Lab6_Task$ gcc writer.c -o writer
student@student-OptiPlex-7000:~/Desktop/Lab6_Task$ gcc reader.c -o reader
student@student-OptiPlex-7000:~/Desktop/Lab6_Task$ ./writer
Enter two numbers: 2
2
Choose operation (1 = Sum, 2 = Product): 1
student@student-OptiPlex-7000:~/Desktop/Lab6_Task$

student@student-OptiPlex-7000:~/Desktop/Lab6_Task
student@student-OptiPlex-7000:~$ cd Desktop
student@student-OptiPlex-7000:~/Desktop$ cd Lab6_Task
student@student-OptiPlex-7000:~/Desktop/Lab6_Task$ ./reader
Result: 4
student@student-OptiPlex-7000:~/Desktop/Lab6_Task$
```